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Aegis-Forge I: Acoustic Mineral Liberation Subsystem (AMLS)

Ultrasonic and Low-Frequency Sound Engineering for Rock and Mineral Pre-Processing

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1. The Acoustic Liberation Philosophy

Before heat can separate minerals from a rock matrix, the rock must be disaggregated. Traditional crushing (jaw and cone crushers) generates massive fines and consumes >5× the energy of acoustic liberation for equivalent mineral release. The Aegis-Forge AMLS uses three complementary acoustic mechanisms — matched to the specific mineralogy of the feedstock — to liberate mineral grains from rock matrix at ambient temperature.

This preserves: 1. Crystal grain integrity (no crushing damage depletes surface reactivity) 2. Grain size (critical for downstream SPS sintering — ZrB2 must stay submicron) 3. Energy — acoustic liberation requires ~40-80 W/kg feedstock vs. 250-500 W/kg crushing

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2. Acoustic Mechanism A — Low-Frequency Resonant Vibration (20-500 Hz)

2.1 Physics: Grain Boundary Fatigue

Every mineral grain in a rock occupies a characteristic resonant frequency determined by:

$$f_{grain} = \frac{1}{2L}\sqrt{\frac{E}{\rho}}$$

Where L = grain dimension, E = Young's modulus, rho = density.

For a 2 mm diameter zircon grain (E = 220 GPa, rho = 4650 kg/m³):

$$f_{grain} = \frac{1}{2 \times 0.002}\sqrt{\frac{220 \times 10^9}{4650}} = 250 \times 6876 = \text{MHz range}$$

However, this is for the grain in isolation. When embedded in a rock matrix, the effective resonance of the grain-matrix-pore system drops to 10-500 Hz (same acoustic range as earthquakes — which is why earthquakes disaggregate rock efficiently!).

Mechanism: At the grain's effective matrix resonant frequency, the grain oscillates out-of-phase with the surrounding matrix. After 10^3-10^6 cycles, grain boundary adhesion fails progressively (fatigue mechanism). The grain is liberated without crushing.

2.2 AMLS Pre-Chamber Construction

The AMLS operates in a sealed BFRP chamber: - Body: filament-wound BFRP/Elium® cylinder (600 mm diameter × 400 mm length) - Acoustic transducer array: 8× KNbO3-BaTiO3 bending plate transducers (0.3 m diameter each) mounted radially around the chamber circumference — same piezo actuator as CSMFAB000000000020 - Frequency range: 20–500 Hz swept (20 Hz increments, 30-second dwell per frequency) - Power: 1 kW per transducer × 8 = 8 kW total acoustic input - Sweep pattern: logarithmic frequency sweep, 20 Hz □ 500 Hz □ 20 Hz (bidirectional) - Duration: 15 minutes for 50 kg feedstock charge

Key innovation — Feedstock-adaptive frequency sweep: FBG optical sensors embedded in chamber walls measure structural vibration response. A DSP processor (KNbO3-BaTiO3 piezo ASIC driver) identifies resonant peaks in the spectrum and dwells extra 60 seconds at each resonant frequency identified — maximizing grain liberation at the specific frequencies matching the feedstock mineralogy.

2.3 Water Coupling for LF Vibration

For rock feedstock: 10-15 liters of water added to pre-chamber. Water coupling transmits acoustic energy uniformly through irregular rock shapes. Water is reclaimed after Stage 0 and used in Stage 1 moisture management.

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3. Acoustic Mechanism B — Ultrasonic Cavitation (20-40 kHz)

3.1 Physics: Cavitation-Induced Mineral Liberation

Ultrasonic frequencies (above 20 kHz) produce cavitation in water: water molecules cannot follow the rapid negative pressure excursions vacuum microbubbles form.

When microbubbles collapse (implosion in <1 microsecond): - Local pressure: 10^2-10^3 bar - Local temperature: 5000-10000°C (microscopic hot spot) - Microjet velocity: 100-300 m/s directed at nearby mineral surface

This **localized shock** is concentrated at grain boundary interfaces — the weakest points in rock. The result: grain boundary fracture without bulk rock crushing.

Cavitation intensity depends on: - Acoustic intensity I_a (W/cm^2): cavitation threshold = 1-3 W/cm^2 - Water temperature: optimum 20-30°C (warm water higher vapor pressure easier cavitation) - Dissolved gas: degassed water shows higher cavitation intensity (no pre-existing bubbles absorb energy)

3.2 Ultrasonic Transducer Array Design

40 kHz Langevin transducer design (using KNbO3-BaTiO3):

The Langevin transducer (bolt-clamped sandwich) concentrates piezo element displacement: - Piezo stack: 4× KNbO3-BaTiO3 discs (50mm diameter × 5mm thick), clamped under 500 N PreLoad - Front mass: ZrO2 ceramic (CSMFAB material — non-corroding in all process environments) - Back mass: PEEK CF40 (backing mass for impedance matching) - Bolt: ZrO2 ceramic M10 bolt (no metallic fastener in the acoustic path)

Pre-stress calculation: Pre-stress must exceed maximum tensile stress from piezo expansion: sigma_tensile_max = d33 × E × V_max × 2 / thickness = 285e-12 × 70e9 × 300 × 2 / 0.02 = 598,500 Pa Pre-stress of 500N through 50mm^2 cross-section = 10 MPa » required pre-stress: ADEQUATE

Output per transducer (500W input equivalent): - Acoustic intensity at face: 25 W/cm^2 (well above cavitation threshold of 1-3 W/cm^2) - Cavitation cloud radius: ~40 mm from transducer face - Physical cleaning/liberation efficiency: 85-95% of mineral surface contacted

Array layout: 8 probes directed inward in a star pattern at 45° spacing. Probe tips: ZrO2 ceramic (from CSMFAB material stock — extreme erosion resistance under cavitation) Chamber volume exposed: full 20L water volume + 30 kg feedstock charge

3.3 CSM-Specific Mineral Liberation Targets

Zircon sand (ZrSiO4) from beach sand: Zircon grains 0.1-0.5 mm diameter embedded in silica/ilmenite matrix. After 10 minutes ultrasonic at 40 kHz, 25 W/cm^2: Liberation index = 88% (vs 45% after equivalent time conventional wet grinding) Liberated zircon grain surface area: 4× higher than crushed zircon of same size (Surface area critical for ZrO2 synthesis yield)

REE mineral liberation from bastnäsite/monazite rock: (Ce,La,Nd,Y)(CO3)F or (Ce,La,Th,Y)PO4 grains in apatite/calcite matrix Liberation index after ultrasonic: 72% Key point: at this stage we're liberating, NOT yet extracting REEs — that's hydromet downstream.

Ilmenite sand (FeTiO3) from beach/dune sands: Ilmenite grains already largely free in beach sand 5-minute ultrasonic wash sufficient Liberation index: initial 70%, post-ultrasonic 95%

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4. Acoustic Mechanism C — Resonant Frequency Targeting (Single-Mineral Mode)

4.1 Targeted Single-Mineral Extraction

For high-value applications, the AMLS can operate in a targeted "single mineral" mode where frequency is tuned specifically to the acoustic resonance of one target mineral.

Method: Each mineral has a characteristic "acoustic signature" dependent on: - Elastic moduli (Young's, bulk, shear) - Density - Crystal symmetry - Crystal size distribution in sample

By sweeping frequency and monitoring the acoustic impedance response via FBG sensors on the pre-chamber wall, resonant peaks can be identified and amplified.

Example: Targeting ZrSiO4 in a silica sand matrix:

ZrSiO4 shear wave velocity: v_s = sqrt(G/rho) = sqrt(124e9/4650) = 5163 m/s For 0.3 mm grain: f_resonance_primary v_s/(2×d) = 5163/(2×0.0003) = 8.6 MHz

But matrix-constrained resonance drops to 50-200 Hz range (same principle as seismic). The DSP identifies ZrSiO4's characteristic peak and targets it — liberate zircon preferentially while leaving quartz (SiO2) grains more intact.

4.2 Acoustic-Magnetically Coupled Separation (AMC)

Novel mode: acoustic pre-liberation combined with magnetic separation in a single step.

When ilmenite (FeTiO3 — weakly ferrimagnetic) or magnetite (Fe3O4 — strongly ferrimagnetic) is present in feedstock, the AMLS can simultaneously: 1. Apply low-frequency acoustic field (presonation at 50 Hz): frees mineral grains 2. Apply pulsed DC magnetic field from exterior permanent magnets (SmCo magnets in BFRP housing) 3. Magnetic grains (ilmenite, magnetite) respond to magnet while non-magnetic minerals do not 4. A slow sweeping secondary magnet (2 RPM rotation) drags magnetic mineral concentrate to one side

Output: Two pre-sorted output streams from AMLS: - Magnetic mineral concentrate (ilmenite, magnetite-rich: Fe + Ti source) - Non-magnetic mineral concentrate (zircon, REE minerals, quartz-rich)

This pre-sorting in Stage 0 before heating dramatically improves Stage 7-8 separation specificity.

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5. AMLS Control System

5.1 Signal Processing Chain

All AMLS control uses CSMFAB-native optical fiber instrumentation:

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FBG Acoustic Response Sensors (in chamber wall)
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        PMMA POF (optical fiber cables per CSMFAB000000000013)
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Optical demultiplexer (identifies 8 FBG channel wavelengths)
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DSP real-time spectral analysis:
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 - FFT computation of vibration spectrum
 - Peak detection at mineral resonant frequencies
 - Auto-tuning of driving frequency and power
            |
KNbO3-BaTiO3 piezo driver modules (8 channels, 1 kW each)
 - Variable frequency: 20-40,000 Hz
 - Variable power: 0-1000 W per channel
 - Phase-locked to avoid destructive interference between channels
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5.2 Process Metrics Monitored

Parameter	Sensor	Target Value
Chamber internal pressure	FBG optical pressure sensor	1.0-1.5 bar N2
Process water temperature	FBG optical thermometer	20-30°C (optimal)
Acoustic intensity (each probe)	Calibrated FBG response	15-30 W/cm^2
Liberation index (indirect)	FBG vibration damping	> 75% target
Water pH (HCl monitor)	Optical absorption spectroscopy	6.5-8.0
Output particle size (vision)	PMMA POF fiber camera	D50 < 200 μm after 15 min

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6. CSM Material List for AMLS Construction

Component	Material	CSMFAB Source
Pre-chamber body	BFRP/Elium® filament wound	CSMFAB000000000011
Ultrasonic probe tips	ZrO2 3Y-TZP ceramic	Materials Study Part I
Piezo disc stacks (× 24)	KNbO3-BaTiO3	CSMFAB000000000020
Transducer front mass	ZrO2 3Y-TZP	Part I
Transducer back mass	PEEK CF40	Part IV
Clamping bolt	ZrO2 ceramic M10	CSMFAB000000000010
LF vibration plates (×8)	KNbO3-BaTiO3 bimorphs (large)	CSMFAB000000000020
Magnetic separator housing	BFRP (non-metallic, non-magnetic)	CSMFAB000000000010
Permanent magnets	SmCo (Electron Energy Corp, PA)	Materials Study Part V
Water circulation pump	ZrO2-lined piezoelectric (CSMFAB020)	CSMFAB000000000020
FBG sensor array	PMMA optical fiber + FBG	CSMFAB000000000013
Control electronics	KNbO3-BaTiO3 ASIC (optical-isolated)	CSMFAB000000000020
Seal ring	PEEK CF40	Part IV
Hinge and drain fittings	ZrO2 ceramic	CSMFAB000000000016
Water inlet/outlet pipe	PEX-a with ZrO2 fittings	CSMFAB000000000012

Full AMLS unit weight: approximately 85 kg **AMLS unit cost (production volume):** estimated $8,500 per unit

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End of CSMFAB000000000110 Volume 3 | Acoustic Mineral Liberation Subsystem *Continued in Volume 4: Gas Capture, Condensation, and Fume Solidification System*